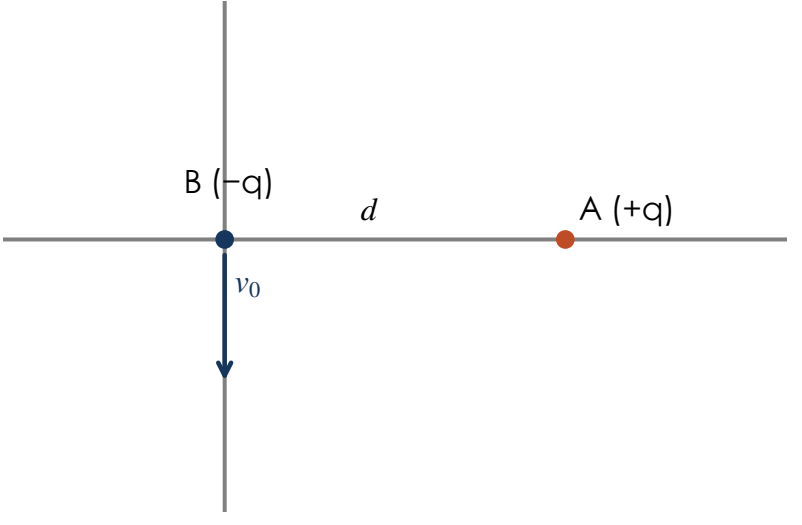相互垂直的无摩擦细管